# Additional file 5: Retained within-site AlphaGenome audit

This incorporated evidence forms part of the replacement manuscript. Citation numbers in this standalone file refer to the bibliography below.

## 1 Design and empirical scope

The initial cohort comprises 120 ClinVar-selected pathogenic or likely pathogenic coding substitutions in *DNM1* (20), *PCCA* (15), *PCCB* (27) and *MMUT* (58) [1]. All three non-reference substitutions were scored at each coordinate. Background profiles were constructed from available genome-wide background variants, with 400 for most scorers but fewer for several. Profiles used dominant maximum-absolute pooling. Euclidean and Poincaré comparisons jointly scaled each site's three profiles and centroid by their maximum norm; the Poincaré transform used fixed curvature $-1$ and radius fraction 0.8. The primary statistic was the mean difference between the selected substitution's distance from the background profile and the larger distance of its two alternatives. It was evaluated for 22 scorers in three geometries with within-site randomization and a declared 66-comparison family. The cosine comparison for the scalar AVI score is undefined and was not interpreted. AVI includes AlphaMissense and consequence-related features [2,3,4].

The replications include 384 held-out primary positions in the four-gene pool and 1,713 primary positions in seven further genes. The four-gene replication followed a locally dated plan; the seven-gene plan was deposited publicly before score retrieval [5]. The initial cohort arose from live ClinVar queries capped at 120 records per gene in descending identifier order, followed by P/LP SNV filtering; its parser excluded intronic records. The pool and panel replications used star-filtered P/LP records from a checksummed ClinVar snapshot. These ascertainment rules limit the population represented by the results. Site-level conditional evidence does not establish performance in an independently sampled gene population.

Permutation tests used 9,999 draws and plus-one p-values [6], with BH correction within each declared analysis family [7]; the original primary analysis also used BY as a dependence-robust check [8]. The expanded-panel missense null restricts pseudo-selection to missense substitutions; its one position with a unique missense substitution contributes a constant. The context-aware null assigns transitions weight 4, corresponding to Ti:Tv 2 before conditioning, and CpG transitions an additional factor of 10, with weights renormalized over eligible substitutions. This specified null produced the reported missense and other-intronic q-values. Earlier cohort consequence analyses and the later baseline and direct-ranker comparisons were post hoc. These analyst-defined mutation weights are sensitivity models, not independently estimated site-specific mutation rates. The highlighted missense and other-intronic mean-statistic q-values derive from zero-exceedance, 9,999-draw Monte Carlo cells. They are adjusted point estimates at the simulation resolution floor; Additional file 3 reports their raw p-values and Monte Carlo precision. The spatial tests in the main article spatial analysis instead use exact finite-label arithmetic.

## 2 Consequence-associated enrichment and surviving signals

In the initial cohort, 17 of 66 primary comparisons were significant after BH correction. For the primary CONTACT_MAPS Poincaré comparison, an exact lattice calculation bounded the uniform-null tail probability at 0.014989–0.015035 and its BH q-value at 0.054961–0.055128. Both bounds leave it above

the threshold. Other statistic and null choices can change this conclusion. These are contact-score profile results, not persistence-diagram results.

The expanded replication gives the more informative consequence contrast. Ranking by AVI Euclidean distance from the background centroid placed the ClinVar allele first at 1,095 of 1,238 nonsense or start-loss positions and 94 of 246 missense positions, approximately 88% and 38%. Under the coding consequence-matched null, 10 of 66 comparisons remained significant, compared with 37 under the uniform null. The coding matched-null analysis restricts pseudo-selection to substitutions sharing the selected allele's consequence and excludes positions with fewer than two eligible substitutions. AVI was therefore evaluated at 452 matchable positions, compared with 1,490 coding positions under the uniform null. The comparison does not isolate a change of null at a fixed set of positions. Ten exceeds the pre-specified bound of one quarter of 37. The planned criterion for calling the coding result mainly consequence class was therefore not met in the retained run. Three post-result redraws of the same Monte Carlo analysis reproduced 37 and 10. Conservative simultaneous 95% Monte Carlo enclosures nevertheless allowed significant-count ranges of 36–37 and 8–11, respectively; these enclosures do not establish the quarter criterion's outcome (Additional file 3).

At the 246 missense positions, AVI's consequence-matched, mutation-aware mean-statistic test remained significant ($q = 0.00132$). The observed mean selected-minus-largest-other Euclidean distance was -0.0128, compared with a simulated matched-null mean of -0.0829. Significance therefore reflects enrichment relative to this null, not a positive average margin. Outside the chr13 genes, its first-rank fraction was 58 of 170, or 0.341; the corresponding observed and simulated-null means were -0.0407 and -0.0785, and q was 0.00165. Additional file 3 records the values and simulation scope. Null-relative enrichment, a positive average margin and preferential first rank are distinct.

At 63 other intronic positions, the three splicing scorers placed the selected allele first at 46, 47 and 47 positions, respectively; each corresponding reported mean-statistic q-value was 0.000252. At 157 canonical splice positions, none of the 63 scorer-by-geometry comparisons was significant. These findings are conditional on the evaluated sites and labels. Missense and splice classifications can incorporate computational evidence related to the evaluated predictors [9,10].

Post hoc, a model-free consequence-severity rule had an expected first-choice fraction of 0.817 under random tie breaking, compared with AVI's observed fraction of 1,192/1,490 (0.800) across all coding positions. This cohort includes 1,238 nonsense or start-loss, 246 missense and six synonymous positions; AVI placed the selected allele first at three of the six synonymous positions. AlphaMissense alone, an input to AVI, also showed a missense preference. These analyses are exploratory and do not provide independent validation.

## 3 Anchors, ranks and the scope of their repair

The sensitivity analyses include mean-rank and top-1 statistics. Replacing the centroid by the zero-effect origin produced the same sixteen significant Euclidean/Poincaré comparisons in its 44-comparison sensitivity family. This supports robustness to that anchor change for those comparisons; it does not establish identical p-values across all analysis families. The historical top-1 analysis designated one maximum using the index returned by `argsort`; it did not require strict superiority or allocate fractional credit for tied maxima. Its mean-rank alternative used midranks. These rules differ from the tie-aware indicator used in the main article spatial analysis. A targeted cache check found no top ties, either

exactly or within absolute tolerance $10^{-12}$, for the reported AVI Euclidean nonsense/start-loss and missense strata, including missense outside chr13. Their three reported counts are unaffected by this distinction. The exploratory CONTACT_MAPS Poincaré top-1 count, 53 of 120, also had no tied maxima; its reported q = 0.0441 belongs to the exploratory 66-comparison rank family and is distinct from the primary mean-statistic result above.

The new direct reference-versus-alternate spatial comparison eliminates an external background centroid by construction. That changes the endpoint. It does not resolve the broader limitations involving consequence, mutation rates, pooling, uncertainty across genes or clinical-label circularity. Ranking alone does not remove an anchor already used to define the ranked distances.

## References


1. Landrum MJ, Chitipiralla S, Kaur K, et al. ClinVar: updates to support classifications of both germline and somatic variants. *Nucleic Acids Research*. 2025;53:D1313–D1321. doi:10.1093/nar/gkae1090.
2. Cheng J, Taylor KR, Nicolaisen L, et al. AlphaGenome Atlas: in silico mutagenesis of the entire human genome improves prioritization and interpretation of non-coding variants. *medRxiv*, preprint, 2026. doi:10.64898/2026.09.16.26363192.
3. Cheng J, Novati G, Pan J, et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science*. 2023;381:eadg7492. doi:10.1126/science.adg7492.
4. McLaren W, Gil L, Hunt SE, et al. The Ensembl Variant Effect Predictor. *Genome Biology*. 2016;17:122. doi:10.1186/s13059-016-0974-4.
5. Iben P. Pre-specified analysis plan: expanded-panel replication of a within-site audit of AlphaGenome Atlas predictions. Zenodo, version 1.0, 2026. doi:10.5281/zenodo.23524782.
6. Phipson B, Smyth GK. Permutation P-values should never be zero: calculating exact P-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*. 2010;9:Article 39. doi:10.2202/1544-6115.1585.
7. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*. 1995;57:289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.
8. Benjamini Y, Yekutieli D. The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*. 2001;29:1165–1188. doi:10.1214/aos/1013699998.
9. Richards S, Aziz N, Bale S, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*. 2015;17:405–424. doi:10.1038/gim.2015.30.
10. Pejaver V, Byrne AB, Feng BJ, et al. Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. *American Journal of Human Genetics*. 2022;109:2163–2177. doi:10.1016/j.ajhg.2022.10.013.